

Ngô Gia Huy

0777499241 | ngohuy270306@gmail.com | linkedin.com/in/ngo-gia-huy-a92336368 | github.com/Platypus27-coder

EDUCATION

Ho Chi Minh City University of Science (HCMUS)

Bachelor of Science in Data Science (GPA: 3.741/4.0)

Ho Chi Minh City, VN

Sep. 2024 – Present

PROJECTS

Offline Speech-to-Speech Translator (VI ↔ EN) | GitHub: [OneVoice](#)

- Built and demonstrated an offline Vietnamese–English speech-to-speech translator on Windows PC, enabling bidirectional voice communication without cloud inference through concurrent ASR, translation, and speech-output workers.
- Fine-tuned **SenseVoice** and **EnViT5** and validated their **ONNX exports** for local inference; integrated the **GIPFormer ONNX baseline** for Vietnamese ASR, domain context and safety routing, and **Windows SAPI** speech output.
- Validated the full pipeline on **4,756 internal noisy and synthetic WAV replay turns**, achieving a **98.28% functional pass rate**. Measured noisy-set ASR WER was **11.32% for Vietnamese** and **0.55% for English**.

Neurosymbolic STEM Reasoning Agent | GitHub: [EXACT_2026](#)

- Built a neurosymbolic hybrid AI agent using **LangGraph** to solve college-level logic and physics STEM problems.
- Designed a hybrid reasoning pipeline routing questions through the **Z3 Theorem Prover** for logic and **SymPy** for physics, with a direct LLM reasoning fallback.
- Implemented an automated code-repair pipeline that detects runtime exceptions and regenerates solver code until successful execution.
- Engineered a RAG pipeline using **Qdrant** and **BAAI/bge-reranker**, deploying the solution via **FastAPI** with a **Qwen-7B** model under 4-bit quantization.

Local RAG Chatbot | GitHub: [RAG_With_ChatUI](#)

- Developed a local document Q&A chatbot using **LangChain** and **Chainlit** for privacy-focused question answering over uploaded PDF and TXT files.
- Built a RAG pipeline using **ChromaDB** for vector storage and **HuggingFaceEmbeddings** on CPU to conserve GPU resources.
- Integrated **Qwen2.5-1.5B-Instruct** in 4-bit quantization with **ConversationalRetrievalChain** and **MMR** search to retrieve diverse context.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL (MySQL), R

LLM and Agentic Systems: Qwen, LangChain, LlamaIndex, LangGraph, RAG, Prompt Engineering, Model Quantization (GGUF, NF4), LoRA/QLoRA, Unsloth

Deep Learning: PyTorch, Transformers, CNN, ResNet, U-Net, Mamba, Hugging Face

Machine Learning and AI: Scikit-learn, Reinforcement Learning, Speech Processing (ASR, TTS)

Data Science and Analytics: Pandas, NumPy, Matplotlib, Seaborn, Vector Databases (ChromaDB, Qdrant)

Software Engineering and MLOps: FastAPI, Docker, Git/GitHub, CI/CD, Airflow, MLflow

CERTIFICATIONS

- VSTEP English Proficiency Certificate:** Score 7.0 (B2).